Data 100 : Bias Variance

$\text{Var}(X) = E[D^2]$ → expectation of deviation squared : $D = X - E[X]$

$= E[(X - E[X])^2]$

$= E[X^2 + 2X E[X] + (E[X])^2]$

$= E[X^2] - 2E[X]E[X] + (E[X])^2$

$= E[X^2] - (E[X])^2$

Similarly

$\text{Var}(X+Y) = E[(X+Y - E(X+Y))^2]$

$= E[(X+Y - E[X] - E[Y])^2]$

$= E[(\underbrace{X - \mu_x}_{a} + \underbrace{Y - \mu_y}_{b})^2]$

$= E[a^2 + b^2 + 2ab]$

$= E[(X-\mu_x)^2 + (Y-\mu_y)^2 + 2(X-\mu_x)(Y-\mu_y)]$

$= E[(X-\mu_x)^2] + E[(Y-\mu_y)^2] + 2E[(X-\mu_x)(Y-\mu_y)]$

$= \text{Var}(X) + \text{Var}(Y) + \underbrace{2E(D_x D_y)}_{\text{Covariance}}$

$\text{Cov}(X,Y) = E[(X - E(X))(Y - E(Y))]$

$= E[XY - XE(Y) - YE(X) + E(Y)E(X)]$

$= E(XY) - E(X)E(Y) - E(Y)E(X) + E(Y)E(X)$

$= E(XY) - E(X)E(Y)$

Since we express

$$r = \frac{\text{Cov}(X,Y)}{\sigma_x \sigma_y}$$

For covariance $= 0$, X & Y are uncorrelated (since $r(X,Y) = 0$)

(But that doesn't make them independent)

or

X & Y are independent

↳ Model Bias : average difference between our predicted value and the true value

$$E[\hat{Y}(x)] - g(x)$$

↳ Model Risk : Mean squared error of prediction

$$E[(Y - \hat{Y}(x))^2]$$

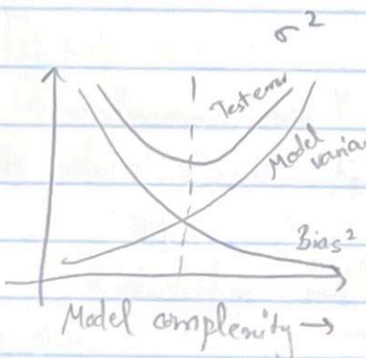
↳ Model Variance : Average of predictions across all possible samples

$$E[(\hat{Y}(x) - E(\hat{Y}(x)))^2]$$

↳ Has got nothing to do with the true model value (Just variations in our own predictions)

Bias-Variance Decomposition:

$$\text{Model Risk} = \text{Observation variance} + \left(\text{model bias}\right)^2 + \text{model variance}$$



PCA

The goal of PCA is to identify the most meaningful basis to re-express a data set

↳ Our hope is that these new basis will filter out the noise and reveal hidden structure i.e. re-expressing the data as a linear combination of its basis vectors

Assumptions:

1) Linearity
2) Large variances have important structure
   (PC's corresponding large variances represent interesting structure)
3) Principal components are orthogonal

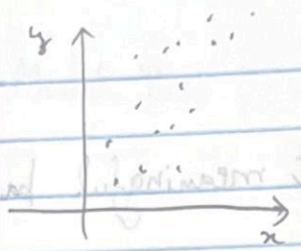
A 2D overview of PCA: Transforms Nxd matrix (X) into an Nxm matrix (Y)

(If d=m, then there is no loss of information)

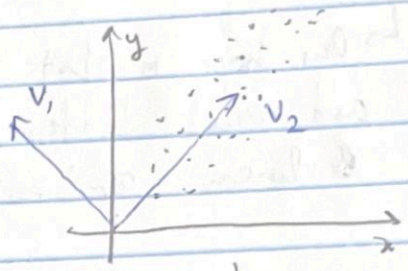
Steps:

1) Centralized data (subtract mean)
2) Calculate dxd covariance matrix (diagonals are variance while off-diagonals are covariance b/w i & j)
3) Calculate the eigenvectors of the covariance matrix (orthonormal)
4) Select m eigenvectors w/ largest eigenvalues

Given we have such data

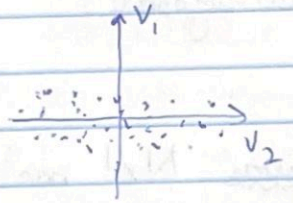


After finding Covariance matrix & using that to get eigenvalues/eigenvectors



V_1 & V_2 are orthonormal vectors that capture the data variance

Now we plot based on new axis

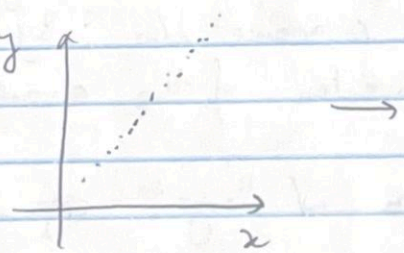


These eigenvectors obtained are called principal components

We see that V_2 captures majority of the data while V_1 captures some of it.

Therefor using only V_2 we can get back projected data

$Y = XV$



Using only PC1, we lose variance in the other direction but retain majority of the structure of the data.

SLR: slope $= r\frac{\sigma_y}{\sigma_x}$; intercept $= \mu_x - \text{slope} \cdot \mu_y$

Linear Regression [w/ changing X, Y parameters]

$\hat{Y} = X\hat{\theta}$

→ Shifting x and y:

- SD of x and y stay the same (Mean changes)
- Parameters don't change (apart from the intercept)
- MSE same
- r same

→ Shifting only x or only y:

- No change in SD (means change)
- MSE same
- Intercept changes

→ Scaled both x and y:

- Everything changes (points move closer or further away to the regression line)

→ Scaling only x

- Intercept stays the same
- MSE same

→ Scaling only y

- Everything changes
- If scaled by $\frac{1}{2}$ → MSE will drop (since predicted values are closer to